Surface Chemistry

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Surface Chemistry

INTRODUCTION :

Surface chemistry is the branch of chemistry in which we deal with action at surfaces or interfaces. Interface or surface are differentiating by separation the bulk phases by a hyphen or a slash i.e. solidgas or solid/ gas. Gases are diffuse rapidly so interface is not possible. It is explained for bulk phase and it is also depending on the particles size of bulk. Various processes like corrosion, electrode processes, heterogeneous catalysis, dissolution and crystallisation are application of interfaces. These type of many industrial applications like analytical work and daily life situations are discussed in this chapter "Surface chemistry".

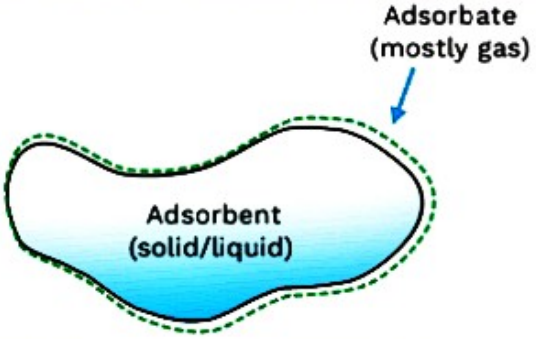
The study of various surfaces and different physical and chemical process taking place on the surface of a solid or liquid.

Adsorption

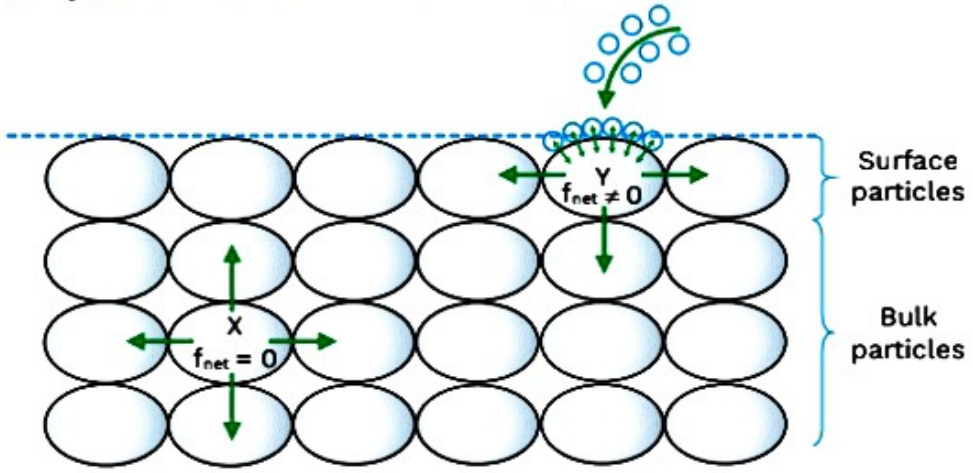
The increase in the concentration of a substance only at the surface of another substance.

The substance accumulated is called adsorbate.

The substance which provides surface is called adsorbent.



e.g. H_2 gas accumulates on Pt electrode (SHE)



In adsorbent, the f_{net} on a bulk particles is zero but for a surface particle it is non-zero. Hence all surface particles have natural tendency to attract and hold adsorbate particles on the surface of adsorbent.

Since it involves attraction so adsorption is an exothermic process.

Due to exothermic, on increase temp., extent of adsorption decrease.

Desorption

It is reverse of adsorption.

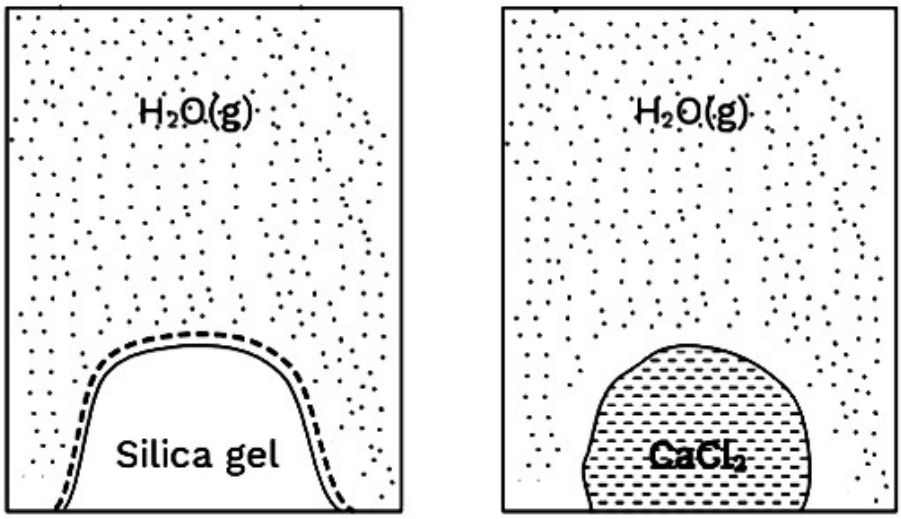
Means removal of adsorbate particle form the surface of adsorbent.

At a constant temperature and Pressure, initially rate of adsorption is max. and decreases with time while rate of desorption is zero and increases with time. So after some time a dynamic Equilibrium is established between adsorption and desorption.

i.e., Adsorbent + Adsorbate $\underset{\text{desorption}}{\overset{\text{Adsorption}}{\rightleftharpoons}}$ Adsorbed

Adsorption is different phenomenon then absorption because:

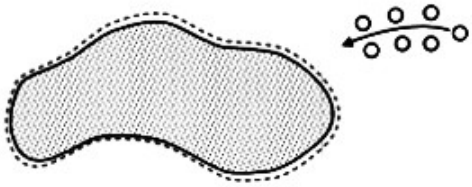
(i) Adsorption takes place only at the surface while ansorption takes place only into the bulk.



Adsorption Absorption

(ii) The rate of adsorption decreases with time while absorption takes place at constant rate.

Sorption:



[It involves both adsorption and absorption]

(Thermodynamic analysis of adsorption)

$\because \boxed{\Delta H_{ads.} < 0}$

Also, during adsorption, since movement of gas particles become restricted, so

$\boxed{(\Delta S)_{ads.} < 0}$

$\because \Delta G = \underset{(-)}{\Delta H} + \underset{(+)}{(-T.\Delta S)}$

We can observe that to keep $\Delta C_p = -ve$ temperature has to below

$\Rightarrow$ Favourable at low temperature.

Adsorption can be classified into two categories namely physical and chemical adsorption.

Physisorption	Chemisorption
Weak Van-der-waal forces between adsorbate and adsorbent	Strong chemical bonds between adsorbate and adsorbent
$\lvert\Delta H_{ads}\rvert$ = 20kJ to 40kJ	$\lvert\Delta H_{ads}\rvert$ = 80kJ to 240kJ
Generally, multilayers are formed	It is a monolayer phenomena
Reversible in nature	Irreversible in nature
Non-specific in nature	Highly specific in nature
Favourable at low temperature	Favourable at relative higher temperature as activation energy is required for reaction

Note: Physisorption convert into chemisorption at higher temp.

Q.1 **Which one is correct statement(s) regarding adsorption of a gas on a solid surface?**
(A) Adsorption is always exothermic
(B) At high temperature Physisorption may transform into chemisorption
(C) As temperature increase, Physisorption increases while chemisorption decreases.
(D) Due to high activation Chemisorption is very slow but Chemisorption is more exothermic than physiorption.

Sol. **(ABD)**

Q.2 **Aqueous solution of Methylene blue, is adsorbed on activated charcoal at 25°. Select the correct statement for this process:**
(A) Activation requires for adsorption at 25°C.
(B) Enthalpy decreases with adsorption.
(C) As temperature increases the adsorption increases.
(D) Nature of adsorption is irreversible.

Sol. **(B)**

Extent of adsorption

It is the measure of accumulated amount of adsorbate (mostly gas) on the surface of adsorbent (solid) which can be given as follow

$$\text{Extent of adsorption}\left(\frac{x}{m}\right) = \frac{\text{mass of gas adsorbed (gm)}}{\text{mass of adsorbent taken (gm)}}$$

For fixed mass of adsorbent taken.

More mass of gas accumulated $\Rightarrow \left(\frac{x}{m}\right)\uparrow$

Factors affecting extent of adsorption

1. **Nature of adsorbate gas**

Among different gases

$$\text{More IMAF} \Rightarrow \left\{\begin{array}{c}\text{more 'a'}\\ \text{or}\\ \text{more '}T_c\text{'}\\ \text{or}\\ \text{easily liquefiable}\\ \text{gas}\end{array}\right\} \Rightarrow \text{more tendency to be adsorbed}$$

[IMAF : Intermolecular attraction force]

2. Nature of adsorbent

They show typical behaviour i.e. suitable adsorbent depends upon the type of gas to be adsorbed.

Gases	Suitable adsorbent
Low IMAF (non-polar)	Transition metal
High IMAF (polar)	Activated charcoal

3. Surface area

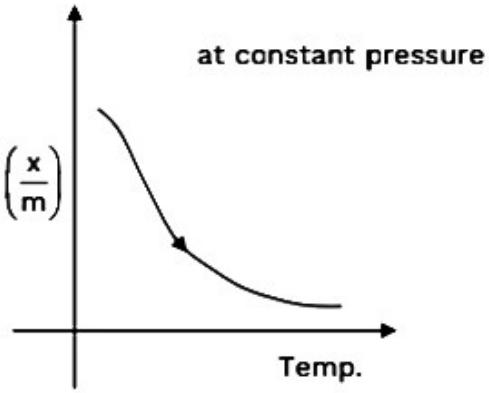
More surface area ⇒ more extent of adsorption

4. Temperature

The variation with temperature is measured at constant pressure hence graphs are known as adsorption isobars

Case-I

For physisorption

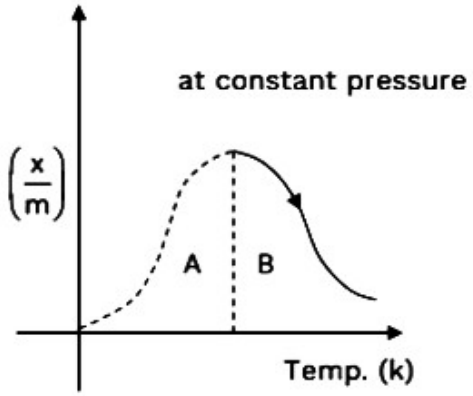


Case-II

For chemisorption

A: on ↑ temp. particle crossing threshold barrier ↑ ⇒ (x/m) ↑

B: on ↑ Temp. (x/m)↓



5. **Pressure :**

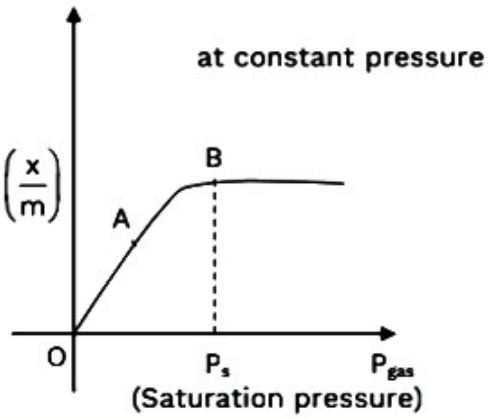
The variation is measured at constant temperature hence graphs are called adsorption isotherms.

Experimentally

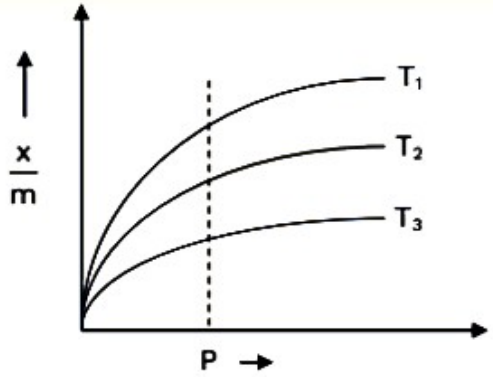
(OA) In low pressure, $(x/m) \propto P^{1}$

(after B) after a certain pressure, $(x/m) \propto P^{0}$

(AB) In modrater pressure, $(x/m) \propto (P)^{\frac{1}{n}}\ (n > 1)$



Q.3



Correct order of T_1, T_2, T_3 ?

Sol. ∵ If more T ⇒ lesser (x/m)

So $T_3 > T_2 > T_1$

Freundlich adsorption isotherm

Before saturation the above curve can be explained by Frendlich Adsorption isotherm equation which is as follow :

$$\left(\frac{x}{m}\right) = k(p)^{\frac{1}{n}}$$

Where, P → pressure of gas

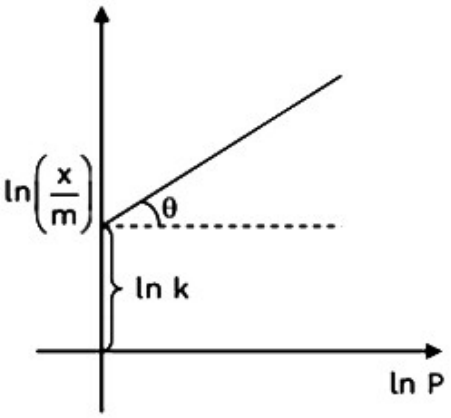
k & n → positive constants $(1 \le n \le \ell)$

(Depend upto gas and adsorbent at a certain temperature)

Also,

$$\ln\left(\frac{x}{m}\right) = \ln k P^{\frac{1}{n}}$$

$$\Rightarrow \underbrace{\ln\left(\frac{x}{m}\right)}_{y} = \frac{1}{n}\underbrace{\ln P}_{x} + \ln k$$



$$\text{Slope} = \left(\frac{1}{n}\right) > 0$$

$$\text{Intercept} = \ln K > 0$$

$$\tan\theta = \left(\frac{1}{n}\right)$$

Langmuir adsorption isotherm :

Freundlich adsorption isotherm

$$\left(\frac{x}{m}\right) \propto (P_s)^{\frac{1}{n}}$$

Note: The above equation is unable to explain the variation in high pressure range so Langmuir equation was introduced which is as follow

$$\boxed{\left(\frac{x}{m}\right) = \frac{a.p}{1 + b.p}}$$

Where P → Pressure Of gas

a, b → positive constants (Not Vander Waals constants)

In Low Pressure	In High Pressire
$P << 1$	$P >> 1$
$(1 + bp) \sim 1$	$(1 + bp) \sim bp$
$\left(\frac{x}{m}\right) \simeq \frac{ap}{1}$	$\left(\frac{x}{m}\right) \simeq \frac{ap}{bp}$
i.e. $\left(\frac{x}{m}\right) \alpha (P)^1$	i.e. $\left(\frac{x}{m}\right) \alpha (P)^0$

Adsorption from Solution

Solids has tendency to adsorb solutes from solutions.

Example: some of the examples are as-

- When solution of aqueous acetic acid (CH3COOH) is mixed with charcoal, some part of the acetic acid is adsorbed by surface of charcoal therefore concentration of acid decreases in the solution.
- When litmus solution shaken with charcoal, it turns colourless.
- In presence of magneson reagent the colour of Mg(OH)2 precipitate changes to blue colour. This is due to adsorption of magneson.

For adsorption, some following observation:

(i) As temperature increases, the amount of adsorption also decreases.

(ii) As surface area of adsorbent increases, the amount of adsorption also increases.

(iii) As the concentration of the solute in solution changes, the extent of adsorption also changes.

(iv) Adsorption also depends on nature of adsorbent as well as adsorbate.

These observations explained by Freundlich's equation by considering pressure difference.

Let's assume the concentration of the solution is taken into account, i.e.

$$\frac{x}{m} = kC^{1/n} \qquad \ldots(3)$$

(C is the equilibrium concentration, i.e., while adsorption is completed).

By taking logarithm of the above expression (3), we have

$$\log\frac{x}{m} = \log k + \frac{1}{n}\log C \qquad \ldots(4)$$

Now Plotting log $\frac{x}{m}$ against log C,

Straight line graph is obtained which is known as freundlich isotherm.

Observation:

At different concentration of acetic acid(CH_3COOH) i.e. Equivalent amounts of charcoal for equal volumes of solutions are added in different flasks and the final concentration for each flask after adsorption is observed. The value of x is calculated with help of change in the initial and final concentrations. Apply above equation and Freundlich isotherm defined.

Applications of Adsorption

There are various application of adsorption. Some of them given below:

(i) Generation of high vacuum:

To generate very high-level vacuum, vacuum pump evacuated the residual adsorbed by charcoal from a vessel.

(ii) Gas masks:

gas mask has activated charcoal surface or mixture of adsorbents which adsorb toxic gases.

(iii) Control of humidity:

To removing moisture and controlling humidity Silica & aluminium gels are utilize as adsorbents.

(iv) Eliminating of colouring matter from solutions:

To removing colour impurities of solution, Animal charcoal used as adsorbate.

(v) Heterogeneous catalysis:

There are several gaseous reactions for industrial importance including solid catalysts in which on the solid surface adsorption of R(reactant) takes place and the rate of reaction(ROR) increases due to catalyst.

Example:

- Haber's process i.e. preparation of ammonia by using iron as a catalyst.
- Preparation of H_2SO_4 with contact process.
- hydrogenation of oils by using finely divided nickel.

These are excellent example of heterogeneous catalysis.

(vi) Segregation of inert gases:

Some of inert gases like argon, krypton and xenon at different temperature, can be removed by adsorption on coconut charcoal. Because of variation in degree of adsorption of gases by charcoal.

(vii) In curing diseases:

Various drugs are used to kill germs by asorption.

(viii) Froth floatation process:

In this process sulphide ore mixed with pine oil and frothing agent and distilled by Segregation it from silica.

(ix) Adsorption indicators:

Precipitate like silver halides have tendency of adsorption of certain dyes like eosin, fluorescein, etc. therefore it will helpful for colour changing at end point and used as indicators.

(x) Chromatographic analysis:

On the basis of adsorption Chromatog is used to finds a number of applications in various analytical and industrial fields.

Q.4 **Select the correct statement, For Freundlich adsorption isotherm with respect to linear plot of log (x/m) versus log p (k and n are constants).**

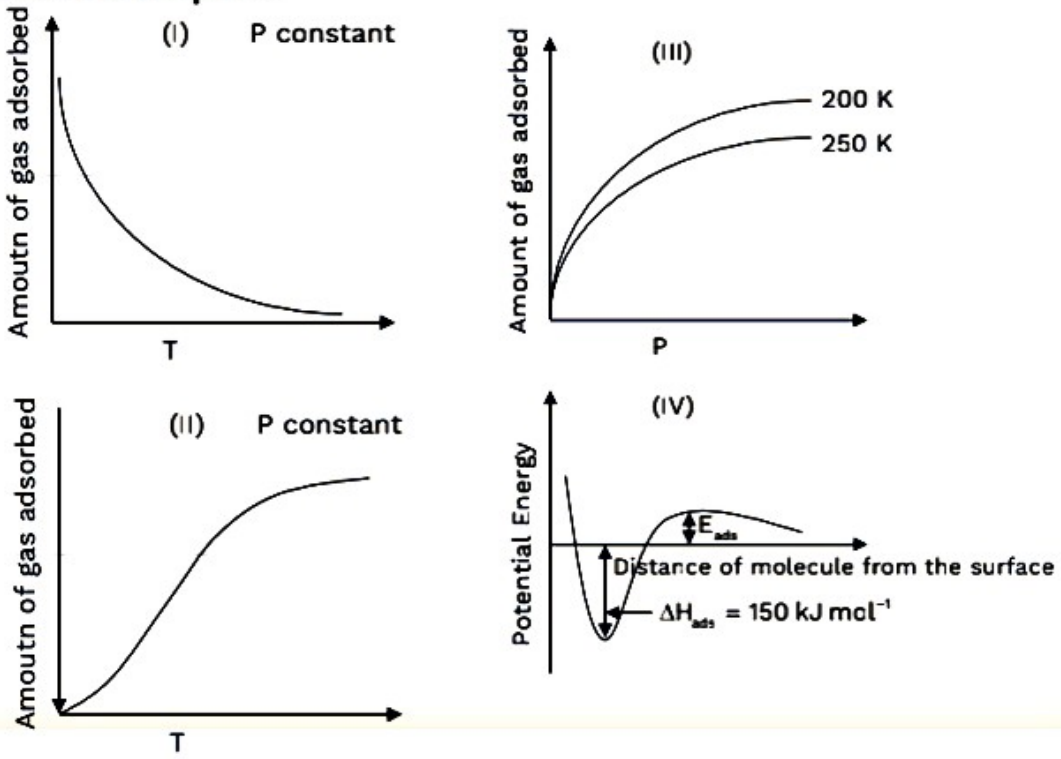
(A) intercept is log (1/n) **(B) Slope are both k and 1/n**

(C) intercept is 1/n **(D) Slope is 1/n**

Sol. **(D)**

$$\underbrace{\log\left(\frac{x}{m}\right)}_{y} = \frac{1}{n}\underbrace{\log P}_{x} + \log k$$

Q.5 **Select correct option regarding I, II, III and IV if under mild condition of temperature and pressure the given curves/data I, II, III and IV represents general trends observed for different adsorption process like physisorption and chemisorption.**



(A) I is related to physisorption and II is related to chemisorption
(B) I is related to physisorption and III is related to chemisorption
(C) IV is related to chemisorption and II is related to chemisorption
(D) IV is related to chemisorption and III is related to chemisorption

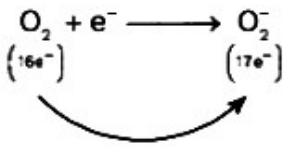
Sol. **(AC)**

(I) → Physisorption
(III) → Physisorption
(II) → Chemisorption
(IV) → Chemisorption

Q.6 **Oxygen is adsorbed on a metallic surface, then transfer of electron from metal to O_2. Correct statement(s) regarding this adsorption is(are)**
(A) physical adsorption of O_2
(B) heat is released
(C) occupancy of π^*_{2p} of O_2 is increased
(D) bond length of O_2 increases

Sol. **(BCD)**

A/C to M.O. configuration upcoming electron will occupy π^*2p_x orbital the bond order will decrease from 2 to 1.5 and hence bond length will increase.



For a binary mixture. If 'd' is the diameter of particle of smaller amount substance, then depending upon the value of 'd' the mixture will be categorised as any of the following

If $d \le 10^{-9}M$	$10^{-9}M < d \le 10^{-6}M$	$10^{-6}M < d \le 10^{-5}M$
True solution	Colloids	Suspension
e.g. sugar is water	e.g. protein in water milk	e.g. sand in water
Homogeneous	heterogeneous	heterogeneous

It means colloid is a heterogeneous system in which one substance (smaller amount) is dispersed into another substance (larger amount)
Smaller amount substance = Dispersed phase (DP)
Larger amount substance = Dispersion medium (DM)
A colloid is also known as Colloidal solution or Sol.

Q.7 **Find the ratio of volume of a colloidal particle (V_c) over the volume of a solute particle in a true solution V_s.**

(A) $\frac{V_c}{V_s} \approx 10^{23}$ **(B)** $\frac{V_c}{V_s} \approx 1$ **(C)** $\frac{V_c}{V_s} \approx 10^{3}$ **(D)** $\frac{V_c}{V_s} \approx 10^{-3}$

Sol. **(C)**

$V \propto r^3$

$V \propto d^3$

$$\frac{V_s}{V_c} = \frac{\left(\leq 10^{-9}M\right)^3}{\left(10^{-9}M < d \leq 10^{-6}M\right)^3}$$

Classification of colloids

I) On the behalf of physical states of DP and DM:

DP	DM	Name of Colloid	Example
Solid	Solid	Solid sol	Gem stones
	Liquid	Sol	Paints
	Gas	Aerosol	Smoke
Liquid	Solid	Gel	Cheese
	Liquid	Emulsion	Milk, Butter
	Gas	Aerosol	Clouds, fog
Gas	Solid	Solid sol	Punice stone
	Liquid	Foam	Whipped cream

Note:

(i) If DM is water then Colloid is known as Aquasol and Hydrosol

(ii) If DM is alcohol then colloid is known as Alcosol.

II) On the basis of interaction between DP and DM:

Lyophillic Colloid	Lyophobic Colloid
Natural attraction between DP & DM	No natural attraction between DP & DM
Can be prepared by simple mixing of DP & DM	Special methods are used for preparation
Naturally stable	Require stabilising agents
Reversible in nature	Pr. in nature

Generally organic sols	Generally inorganic sols
e.g. starch, protein, cellulose, gelatin etc.	e.g. sols of metals, oxides, sulphides etc.

(III) On the basis of charge:

In sol. DP particles carry either positive or negative charge and the DM has an equal and opposite charge.

Depending upon the polarity of DP particles, there are two types.

Positive sol	Negative sol
DP → positive DM → negative	DP → negative DM → positive
e.g. metal oxide sols, metal hydroxide sols, Basic dye (methylene blue) Haemoglobin etc.	e.g. metal sulphides sols sols of metals acidic dye Gum solution etc.

(IV) On the basis of type of DP particle:

(i) Macromolecular colloid

In such sols, one DP particle is actually a single particle having size already of colloidal range.

e.g. Starch, protein, cellulose etc.

(ii) Multimolecular colloids

In such sols, one DP particle is actually aggregation of large number of small atoms or molecules

e.g. In sulphur sol, $(d)_{S\text{-}atom} < 10^{-9}$ m

but

1 DP-particle = Collection of nearly 1000 s-atoms

Associated colloids (Micelles)

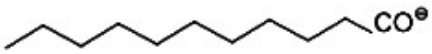
Some electrolytes furnish either cation or anion having very long carbon chain, when dissolved in water.

Such ions are called as surfactants.

Example :

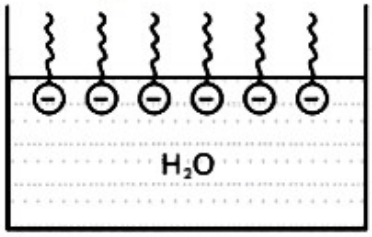
Sodium stearate $(C_{17}H_{35}COO^{-}Na^{+})$

surfactant



Hydrophobic tail(non – polar) Hydrophillic head(polar)

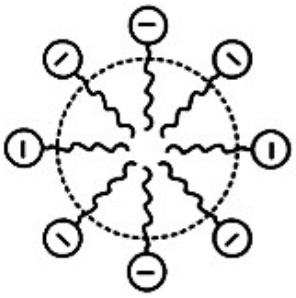
When these types of ions are present in water in low concentration, they are arranged as follow :



at low concentration

As concentration increases, these hydrophilic heads repel each other due to which their hydrophobic tails come closer forming a group of nearly 100 such ions in a spherical way which is called Micelle.

After a certain higher concentration :



For micelle formation :

(i) [Surfactant] $\geq$ CMC (Critical micelle concentration)

(ii) Temperature $\geq T_k$ (Kraft's temperature)

i.e., Favourable at high temperature :

So, $\Delta H > 0$

$S > 0$

Note :

If length of surfactant ion is larger than required CMC will lesser and vice-versa.

Q.8 **S-1 : The formation of Micelles by surfactant molecules possible above the critical micellar concentration (CMC).**
S-2 : At CMC the conductivity decreases of a solution having surfactant molecules.
(A) S-1 is true, S-2 is true; S-2 is a correct explanation for S-1.
(B) S-1 is true, S-2 is true; S-2 is not a correct explanation for S-1.
(C) S-1 is true, S-2 is false.
(D) S-1 is false, S-2 is true.

Sol. **(B)**

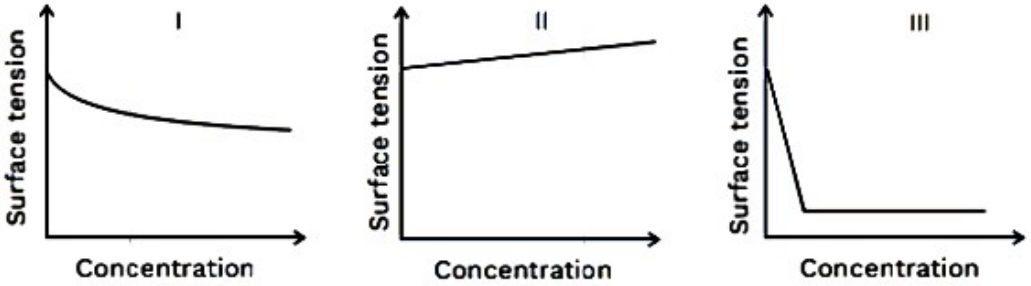
Q.9 **Select the species that will produce micelles in aqueous solution at the lowest molar concentration?**

(A) $CH_3(CH_2)_{15}N^+(CH_3)_3Br^-$ **(B)** $CH_3(CH_2)_{11}OSO_3^-Na^+$

(C) $CH_3(CH_2)_6COO^-Na^+$ **(D)** $CH_3(CH_2)_{11}N^+(CH_3)_3Br^-$

Sol. **(A)**

Q.10 **Select the correct statement about given graphs. If the qualitative graphs (I), (II) and (III) given below, As the surface tension changes with molar concentration of three different aqueous solution of KCl, CH_3OH and $CH_3(CH_2)_{11}OSO_3^-Na^+$ at room temperature.**



(A) I : KCl II : CH_3OH III : $CH_3(CH_2)_{11}OSO_3^-Na^+$

(B) I : $CH_3(CH_2)_{11}OSO_3^-Na^+$ II : CH_3OH III : KCl

(C) I : KCl II : $CH_3(CH_2)_{11}OSO_3^-Na^+$ III : CH_3OH

(D) I : CH_3OH II : KCl III : $CH_3(CH_2)_{11}OSO_3^-Na^+$

Sol. **(D)**

S.T. $\propto$ (IMAF at surface)

Preparation of Colloids

Lyophilic colloids can be prepared by simple mixing of DP and DM.

To prepare lyophobic solution, special methods are used which can be divided into following two categories :

(i) Dispersion methods :

In these methods, particles of big sizes are broken in particles of colloidal range.

(ii) Condensation methods :

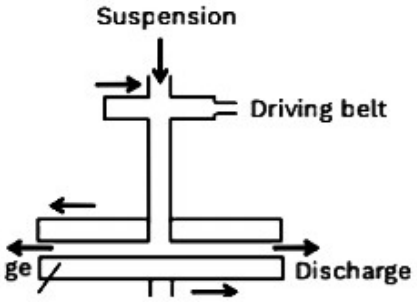
Particles of small sizes marge together to from particles of colloidal range.

Methods of Dispersion

(i) Mechanical dispersion :

• Firstly, Solid substance is finely ground by specific methods. • These substance combined with dispersion medium which form suspension.

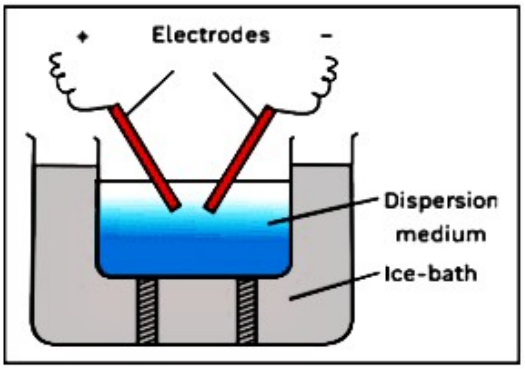
- Suspension is mixed with colloid mill.(colloid mill simplest form of 2 metal discs held at a small distance).
- By this process particles size comes in range of colloidal and dispersed.
- In this solution Stabilizer is also used for stabilize the solution.
- Colloidal graphite and printing ink (lubricants) Tannin is considered as a stabilizer in the colloidal graphite and gum arabic in lampblack colloidal solution (Indian ink).



(ii) Electrolytic-dispersion(Bredig's arc method):

Colloidal solution of metals like gold(Au), silver(Ag), platinum(Pt), etc are formed by bredig's arc method.

- An arc is generated by metal electrodes.
- Container placed in an ice bath.
- Heat generated by arc, vaporised the several metals which further condensed under cold water.
- Strong electrolyte like KOH is added in water to stabilized colloidal solutions.



(iii) Ultrasonic dispersion :

Ultrasonic waves like sound waves (high frequency wave) can be generated by quartz crystal discs are which are linked with a high frequency generator. Ultrasonic waves are used preparation of colloidal solutions. Dispersion is easy with the help of ultrasonic waves of various substance like metal oxides, sulphur, mercury, etc.

(iv) Peptization Process :

The process of converting ppt. into a solution by shaking the ppt. with DM in presence of an electrolyte.

Fresh ppt + DM + Small amount of strong electrolyte ⇒ Solution

The strong electrolyte used for stability is known as peptizing agent.

The strong electrolyte should furnish one similar ion as that in ppt particle.

This common ion provides stability to the solution by its selective adsorption on the surface of ppt particle.

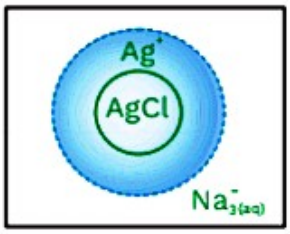
Example : Formation of AgCl sol.

Case I : by NaCl



$AgCl.Cl^-$ solution

Case II : by $AgNO_3$



$AgCl.Ag^+$ solution

Purification of colloids :

Purification process of Colloidal Solutions

In Colloidal solutions generally contains excess amount of electrolytes and small amount of other soluble impurities. To stabilize colloidal solution traces of electrolyte is essential, which useful for more quantities coagulate it. To decrees the concentration of soluble impurities to a requisite min this method is used and this method is called as purification of colloidal solution. Various steps involve in the purification of colloidal solution:

(i) Dialysis
(ii) Electro-dialysis
(iii) Ultrafiltration
(iv) Ultra-centrifugation

(i) Dialysis:

This is a method is used for remove dissolved impurity from a colloidal solution by taking appropriate membrane which is based on diffusion. Particles like



small molecules or ions (solvent) in a true solution can pass through animal membrane (bladder) or cellophane sheet while colloidal particles do not. So basically animal membrane can be utilized for dialysis and the apparatus which used for dialysis is known as dialyser.

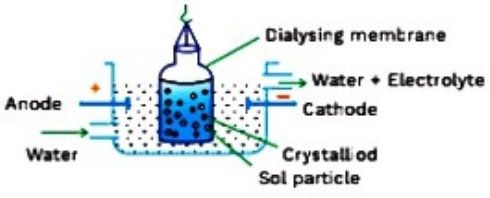
Procedure:

- A bag of suitable membrane is suspended in a vessel (containing colloidal solution) in which passing fresh water is continuously showing in figure.
- Due to diffusion of the molecules and ions they collected with outer water and pure colloidal solution is remaining

lis causing bacteria, spirochete but is also toxic to humans.

(ii) Electro-dialysis:

Dialysis is quite slow process. We can be faster dialysis by applying an electric field, if the soluble substance in the impure colloidal solution are only an electrolyte. Then dialysis process in presence of electric field is called as electro dialysis.



- This colloidal solution is put in a bag having suitable membrane while pure water is taken outside.
- As shown in figure electrodes are arrange and connected with battery.
- Due to electricity ions present in the colloidal

solution migrate out to the oppositely charged electrodes.

(iii) Ultrafiltration:

Ultrafiltration is separation method of colloidal particles from the solvent and soluble solutes present in the colloidal solution by special filters, which are permeable to all substances except colloidal particles.

- Due to too large pores colloidal particles can pass through ordinary filter paper. So that the size of pores in filter paper can be reduced in size by impregnating with collodion solution to stop the flow of colloidal particles. The normal collodion is a 4% solution of nitro-cellulose in a mixture of alcohol and ether.
- Ultra-filter paper, used for the colloidal particles to split from rest of the materials. It is a slow method. So it speeds up the procedure, pressure or suction is used.
- The colloidal particle remains on the ultra-filter paper are and to get a pure colloidal solution stirred with fresh dispersion medium (solvent).

(iv) Ultra-centrifugation:

Ultracentrifugation requires the separation of colloidal particles from the impurities by centrifugal force.

Procedure:

- The impure sol is carried in a tube and the tube which is placed in an ultra-centrifuge and it is rotated at very high speeds.
- The colloidal particles left at the bottom of the tube and the impurities remain in the solution. This solution is termed as centrifugation.
- The colloidal particles are separate out from the tube and mixed with an appropriate dispersing medium. Finally the pure sol is obtained.

Adsorption and Desorption

Q.11 **Gas masks containing activated charcoal to remove poisonous gases from atmosphere acts on principle of**

(A) Adsorption **(B) Absorption**
(C) sorption **(D) All of these**

Sol. **(A)**

Charcoal adsorbs poisonous gases from atmosphere preventing the person wearing mark from exposure to these gases.

Q.12 **Which forms multimolecular layer during adsorption**

(A) Physical adsorption **(B) Chemisorptions**
(C) Both (A) and (B) **(D) None of these**

Sol. **(A)**

Physical absorption forms multimolecular layer due to diffusion through poses & establishing weak force wirte solid.

Q.13 **Physical adsorption turns in to desorption when –**

(A) increase in temperature **(B) decrease in temperature**
(C) increase in pressure **(D) increase in concentration**

Sol. **(A)**

High temperature weakens the bond between adsorbate & adsorbent.

Factors affecting adsorption and catalyst

Q.14 **Catalysts are more effective in**

(A) Finely powdered state **(B) Colloidal state**
(C) Rough surface **(D) All of these**

Sol. **(A)**

Catalysts are more effective in powdered form as it provides more surface area.

Q.15 **Which gas is strongly adsorbed by charcoal**

(A) CO **(B) N_2** **(C) H_2** **(D) NH_3**

Sol. **(D)**

More easily liquefiable gases (NH_3) are readily adsorbed by charcoal

Q.16 **Select statement correct for catalyst –**

(A) A catalyst remain unchanged in the end of chemical reaction.

(B) A catalyst react in a chemical reaction

(C) All kinds of catalysts act as catalyst poisoning

(D) A catalyst structure changes at the end of the reaction

Sol. **(A)**

A catalyst remain unchanged chemically at the end of chemical reaction.

Q.17 **Physical adsorption is accompanied by**

(A) Decrease in entropy of system **(B) Decrease in enthalpy**

(C) The value of $T\Delta S$ is negative **(D) All of these**

Sol. **(D)**

Due to adsorption, decrease in enthalpy as well as decrease in entropy of the system

Freundlich & Langnuir adsorption isotherm

Q.18 **Select correct regarding slope For adsorption of gas on solid surface when the plots of log x/m vs log P is linear–**

(A) K **(B) log K**

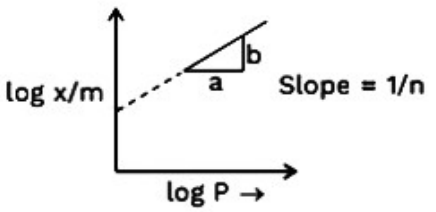
(C) log C **(D) $\frac{1}{n}$ (n being integer)**

Sol. **(D)**

$x / m = k.p\frac{1}{n}$

$\log x/m = \log k + \frac{1}{n}\log P$

Slope $= \frac{1}{n}$ where n being integral.



Q.19 **Langmuir adsorption isotherm is best suitable for**

(A) Chemisorptions **(B) physisorption**

(C) both **(D) none**

Sol. **(A)**

Langmuir adsorption isotherm – deals with monomolecular layer possible for chemisorption

Q.20 **Correct equation of Langmuir adsorption isotherm is**

(A) $\frac{x}{m} = \frac{ap}{1+bp}$ **(B)** $\frac{x}{m} = ap^n$ **(C)** $\frac{x}{m} = kC^{1/n}$ **(D) none of these**

Sol. **(A)**

Langmuir adsorption isotherm

$$\frac{x}{m} = \frac{ap}{1+bp}$$

$$\frac{m}{x} = \frac{1}{ap} + \frac{bp}{ap} \qquad \left(\therefore \frac{1}{ap} \to 0\right)$$

$$\frac{m}{x} = \frac{b}{a}$$

$$\frac{x}{m} = \frac{a}{b}$$

Colloidal solution and its application

Q.21 **The number of phases in colloidal system are**

(A) One **(B) Two** **(C) Three** **(D) Four**

Sol. **(B)**

Two phases in colloidal system

(a) Dispersed phase (b) Dispersion medium

Q.22 **A dispersion of AgCl in water is**

(A) Hydrophilic colloid **(B) An emulsion**

(C) An alcosol **(D) Hydrophobic sol.**

Sol. **(D)**

Lyophobic sols include sols. of metals, and their insoluble compounds like sulphides, chloride and oxides. They need stabilizing substances for preservation, dispersion medium is water.

Q.23 **Fog is an example of**

(A) Solid dispersed in gas **(B) Liquid dispersed in gas**

(C) Liquid dispersed in solid **(D) Solid dispersed in liquid**

Sol. **(B)**

Fog is an example of liquid dispersed in gas i.e. moisture (liquid droplets) is dispersed in air.

Q.24 **Which of the following constitute irreversible colloidal system in water as dispersion medium**

(A) Clay **(B) Platinum** **(C) $Fe(OH)_3$** **(D) All of these**

Sol. **(D)**

Lyophobic sols. are less stable on evaporation of solvent, residue can not be easily transformed back into colloidal state by ordinary means lyopholic colloids are also called irreversible colloids like clay, platinum, $Fe(OH)_3$.

Q.25 **Which one among the following sols is hydrophobic**

(A) Gum **(B) Gelatin** **(C) Starch** **(D) Sulphur**

Sol. **(D)**

Colloidal solutions in which the dispersed phase has very little affinity for the dispersion phase are termed as lyophobic (solvent hating) like – sulphur-sol (less stable). When water is used as solvent, sol is called hydrophobic

Q.26 **Point out the false statement**
(A) Colloidal sols are homogenous
(B) Colloids carry +ve or -ve charge
(C) Colloids show tyndall effect
(D) The size range of colloidal particles is 10 – 1000Å

Sol. **(A)**

Colloidal sols. are homogeneous is wrong statement colloidal sols. are not classified in homogenous and heterogeneous.

Q.27 **A liquid is passed through the filter paper and it scatter a beam of light but leaves no residue.**
(A) A suspension **(B) Oil** **(C) A colloidal sol** **(D) True solution**

Sol. **(C)**

Separation with filter paper not possible for colloidal solution.

Q.28 **Which of the following is a hydrophilic colloidal sol.**
(A) Barium sulphate sol. **(B) Arsenious sulphide sol.**
(C) Starch sol. **(D) Silver iodide so.**

Sol. **(C)**

Starch sol. is example of hydrophilic sol.

Q.29 **In which of following system the dispersed phase and dispersion medium are both solid**
(A) Foam **(B) Dust storm** **(C) Coloured glass** **(D) Paints**

Sol. **(C)**

Solid sol – Ruby glass (coloured glass)

Miscelles, Associated colloids, Preparation of colloidal solution and It's purification

Q.30 **The process of separation of colloids by passing through semi permeable membrane is called**

(A) Filteration **(B) Electrophoresis**

(C) Dialysis **(D) Ultra filteration**

Sol. **[C]**

Dialysis → Dialysis if separation of colloids by passing through semi permeable membrane.

Q.31 **The redispersal of a freshly precipitated substance into a sol by the addition of an electrolyte in common is known as**

(A) Aggregation **(B) Condensation** **(C) Coagulation** **(D) Peptization**

Sol. **(D)**

It is definition of peptization.

Q.32 **Which of the following chemical equation represents the formation of colloidal solution**

(A) $Cu + CuCl_2 \rightarrow Cu_2Cl_2$ **(B)** $2Mg + CO_2 \rightarrow 2MgO + C$

(C) $2HNO_3 + 3H_2S \rightarrow 3S + 4H_2O + 2NO$ **(D) Both (B) and (C)**

Sol. **(C)**

Sulphur sol can be obtained when H_2S is bubbled through an oxidizing agent (nitric acid)

$2HNO_3 + 3H_2S \rightarrow \underline{3S} + 4H_2O + 2NO$

Properties of colloidal solution and it's stability

Q.33 **Select correct option for which Tyndall effect is not observed**

(A) Suspensions **(B) Emulsions** **(C) Sugar solution** **(D) Gold sol.**

Sol. **(C)**

True solution does not show tyndall effect like sugar solution.

Q.34 **The average molecular mass of colloidal can be determined by**

(A) Tyndall effect **(B) Boiling of colloidal**

(C) Osmotic pressure measurement **(D) Flocculation value**

Sol. **[C]**

From osmotic pressure measurement, we can determined average molecular mass of colloids

Q.35 **Which of the following property is the Brownian movement of colloidal sol.**

(A) Electrical **(B) Optical** **(C) Mechanical** **(D) Colligative**

Sol. **(C)**

It is creaseless zigzag motion of colloidal particles. So mechanical property.

Q.36 **Which one is not represented by sols**

(A) Absorption **(B) Tyndall effect**

(C) Flocculation **(D) Paramagnetism**

Sol. **(D)**

Adsorption, tyndall effect and flocculation are properly of sols.

Q.37 **The stability of hydrophobicsol is due to**

(A) Solvation of colloidal particles

(B) The charge on the colloidal particles

(C) The size of the particles

(D) None of the above

Sol. **(B)**

The particles carry a characteristic charge either positive or negative.

Q.38 **As_2S_3 sol is a**

(A) Neutral colloid **(B) Negative colloid**

(C) Positive colloid **(D) None of the above**

Sol. **(B)**

As_2S_3 sol. is negative colloid because this sol. may be partially or completely precipitated.

Q.39 **The presence of colloidal particles of dust in air imparts blue colour to the sky. This is due to**

(A) Absorption of light **(B) Reflection of light**
(C) Refraction of light **(D) Scattering light**

Sol. **(D)**

Scattering of light reason for blue colour for sky.

Q.40 **Select the correct substance which gives a positively charged sol –**

(A) Gold **(B) A metal sulphide**
(C) Ferric hydroxide **(D) An acidic dye**

Sol. **(C)**

In $FeCl_3$ is added to excess of hot water positively charged sol of ferric hydroxide is formed due to adsorption of Fe^{+3} ions.

$Fe(OH)_3/Fe^{+3}$

Positively charged sol.

Q.41 **The stability of the dispersed phase in a lyophobic colloids is due to**

(A) High viscosity of the medium
(B) The formation of electrical layer between two phases
(C) High surface tension of sol
(D) None of these

Sol. **(B)**

There is adsorption of common ion on the colloidal sol and an electrical double layer is formed. Thus, dispersed phase in a lyophobic colloidal sol is stabilized.

Q.42 **Tyndall effect is shown by**

(A) Colloidal solution **(B) Osmotic solution**
(C) Isotonic solution **(D) Hypertonic solution**

Sol. **(A)**

Only colloidal solution due to large of particles of colloids.

Coagulation of sols (Hardy Schultz rule)

Q.43 **Which one of the following will have the highest coagulating power for a ferric hydroxide solution**

(A) NaCl **(B) $BaCl_2$** **(C) K_2CrO_4** **(D) $K_3[Fe(CN)_6]$**

Sol. **(D)**

The power of Coagulation of an electrolyte is directly proportional to the fourth power of the valency of the active ions.

Fe^{+3} has highest valency.

Q.44 **Ferric chloride is applied to stop bleeding because**

(A) Fe^{3+} ions coagulate blood which is a negatively charged sol

(B) Cl^- ions cause co-agulation of blood

(C) $FeCl_3$ react with the constituents of blood

(D) Blood absorbs $FeCl_3$

Sol. **(A)**

Fe^{+3} ion has greater tendency (Coagulating power) which coagulated blood.

Q.45 **In case ion coagulation of a given colloidal solution depends upon**

(A) Its size of ion **(B) The magnitude of charge**

(C) The sign of charge **(D) Both magnitude and sign of charge**

Sol. **(D)**

The ability of ion to bring about coagulation of a given colloidal solution depends upon both magnitude and sign of charge.

Q.46 **An arsenious sulphide sol carries a negative charge. Which of the following maximum precipitating power for this sol?**

(A) K_2SO_4 **(B) $CaCl_2$** **(C) Na_3PO_4** **(D) $AlCl_3$**

Sol. **(D)**

Higher the valency maximum precipitating power for this sol.

Q.47 **A sol has positively charged colloidal particles. Which of the following solutions is necessary required in lowest concentration for co-agulation**

(A) NaCl **(B) $K_4[Fe(CN)_6]$** **(C) $ZnCl_2$** **(D) Na_2SO_4**

Sol. **(B)**

Higher the coagulatating power of ion due to higher valency so decreases the concentration for ion.

$Fe(CN)_6)^{-4}$ has lowest concentration for co-agulation.

Protective colloids and gold number

Q.48 **Gelatin protects**

(A) Gold sol. **(B) As_2S_3 sol.** **(C) $Fe(OH)_3$ sol.** **(D) All**

Sol. **(D)**

Lower the gold number, higher will be the protective power so gelatin have low range (gold number 0.005 – 0.01) so have maximum protective power.

Q.49 **Which of the following method is used for sol. destruction**

(A) Condensation **(B) Dialysis**

(C) Diffusion through animal membrane **(D) Addition of an electrolyte**

Sol. **(D)**

Addition of electrolyte coagulates.

Q.50 **Purple of cassius is**

(A) Colloidal solution of gold **(B) Colloidal solution of silver**

(C) Colloidal solution of platinum **(D) Oxyacids of gold**

Sol. **(A)**

Colloidal solution of gold is called purple of cassius.